

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title: COMPLETED RESEARCH		Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 9

(1) Research Program/Project/Study Title: Utilization and Evaluation of Sweetpotato (<i>Ipomea batatas</i>) as Filler in Meat Loaf Formulation	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	6
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Food/sweetpotato		(8) Discipline/Field of Specialization Food Science	

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 9

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Fillers and additives are made from meat and non-meat products. They are added to increase the bulk of the food, to extend shelf life, and to improve flavor and texture. This study was conducted to determine the acceptability of meat loaf as affected by varying levels of sweetpotato in terms of color, aroma, taste, texture, flavor, and general acceptability; to determine the physico-chemical characteristic of meatloaf with varying levels of sweetpotato as filler; to evaluate the browning and yellowness indices of the finished product; to assess consumer acceptability of the formulated product and to determine the production cost of meatloaf as affected varying levels of sweetpotato. The study was composed of six (6) treatments with sweetpotato levels ranging from 0 to 125 grams. The finished product was subjected to sensory evaluation, color analysis, physico-chemical analysis, consumer testing, and cost analysis. Results of the study showed that the highest level of sweetpotato obtained the highest acceptability rating in terms of color, aroma, taste, texture, flavor and general acceptability with ratings of 7.44, 7.49, 7.56, 7.49, 7.38 and 7.93 respectively. Likewise, physico-chemical analysis showed that the highest level of sweetpotato in the formulation resulted in the highest total soluble solids (TSS) of 21°B and the lowest pH of 6.69. Similarly, the highest level of sweetpotato resulted the lowest browning index, highest yellowness index, and lowest production cost. The meatloaf formulation with the highest level of sweetpotato was liked by 84% of consumers while only 16% disliked the product. Generally, the highest level of sweetpotato in meatloaf formulation resulted in a lower production cost and increased volume of production.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Fillers and additives are made from meat and non-meat products. They are added to increase the bulk of the food, to extend shelf life, and to improve flavor and texture. Fillers include soy concentrate, flour, starches, breadcrumbs, and Mechanically De-boned Meats – such as chicken and turkey parts – materials, separated as the remaining meat on bones. Fillers were originally used to reduce costs but they were later used to make meat products healthier by adding dietary fiber; or to improve the texture. Meatloaf is a dish of ground meat either pork or beef that has been combined with other ingredients, including fillers and extenders, then formed into the shape of a loaf.

With the onset of the African Swine Fever (ASF), the growth rate of pork production, dramatically decreased by 20.8% in 2021. In 2023, pork production decreased from 1,000,000 to 925,000 MT in response to the continued spread of ASF into new provinces within the Central and Western Visayas regions. As a consequence, ASF led to higher pork prices, impacting vulnerable consumers' livelihoods and affecting farmers. ASF is a highly contagious viral disease affecting both domestic pigs and wild boars. ASF outbreak significantly impacted the Philippine pork industry, resulting in the culling of over 300,000 pigs.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 3 of 9

On the other hand, sweetpotatoes are a staple food in many parts of the world. They are a good source of fiber, potassium, vitamins, and other essential nutrients. It's rich in antioxidants called beta carotene, which is very effective at raising blood levels of vitamin A, particularly in children. Sweetpotatoes are nutritious, high in fiber, very filling, and delicious.

The closest prior arts to this study are US3952111A which disclosed that meat extender is derived from wheat kernels and is added to meats such as beef, pork, lamb, fowl and fish to increase the supply thereof. The kernels are boiled and dried to a moisture content of about 20%.

Another prior art has patent number US6419977 which comes from the United States. The invention relates to nutritional meat extender compositions which have been modified to alter the texture or grain of meat preparations while at the same time fortifying the meat with additional nutrients for patients in need of same.

Considering the current situation of the swine industry in the country, the need to add fillers into meat products formulation especially those that used pork as one of the main ingredients. Thus, extension of meat and meat-products with sweetpotato will not only reduce production costs and improve its nutritional quality but also help solve the problem of pork shortage brought by the ASF. Hence, this study was conducted.

Objectives of the Study

Generally, this study aimed to develop and evaluate the quality of meatloaf as affected by varying levels of sweetpotato.

The specific objectives are:

1. To determine the acceptability of meat loaf as affected by varying levels of sweetpotato in terms of color, aroma, taste, texture, flavor and general acceptability;
2. To determine the physico-chemical characteristic of meatloaf with varying levels of sweetpotato as filler;
3. To evaluate the browning and yellowness indices of the finished product;
4. To assess consumer acceptability of the formulated product.
5. To determine the production cost of meatloaf as affected by varying levels of sweetpotato.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Materials

The materials and tools used in the conduct of the study were the following: mixing bowls, chopping board, metal tray, knife, pot, frying pan, meat grinder, digital weighing scale, yellow colored- sweet potato, potable water, ground pork, slotted spoon, autoclave, blender and glass jars.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 4 of 9

Experimental Treatment

The study was composed of six (6) experimental treatments, namely: T0- 0g sweetpotato, T1- 25g sweetpotato, T2- 50g sweetpotato, T3- 75g sweetpotato, T4- 100g sweetpotato T5- 125g sweetpotato

Meat loaf Formulation and Preparation

Meatloaf formulation followed the ingredients and procedure of beef loaf except substituting beef with pork. The ingredients include the following: Stuffing mixture (Mashed sweetpotato), Ground Meat, Onion, pepper, milk, egg, ketchup, Worcestershire sauce, paprika, salt, oregano, ground mustard, Phosphate, Monosodium Glutamate, and Prague powder.

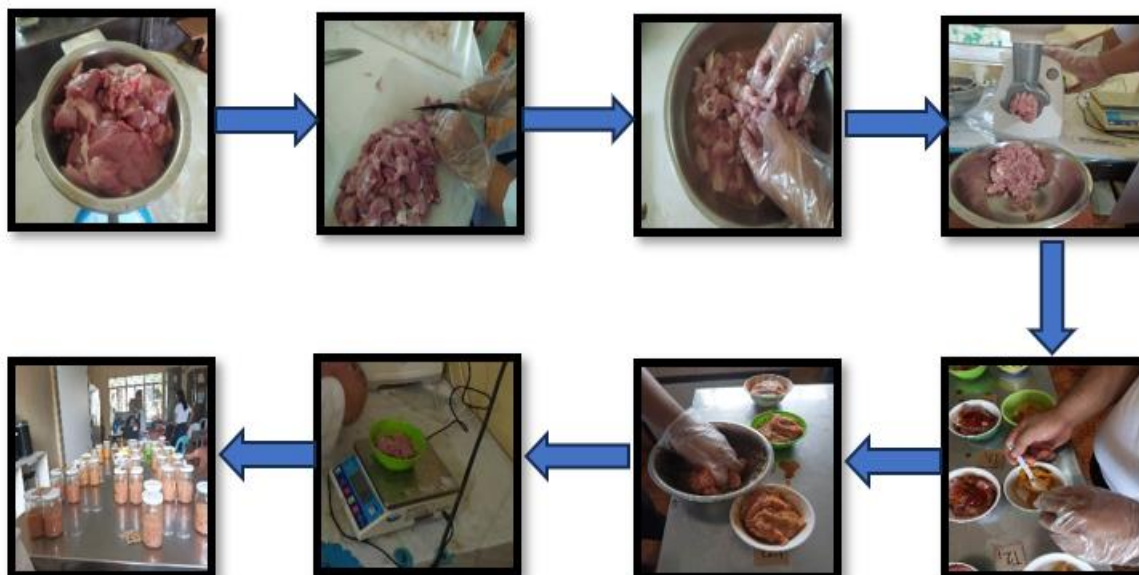


Figure 1. Process flow of meat loaf formulation

Sensory Evaluation

Meat loaf with varying levels of sweetpotato as extender was subjected to sensory evaluation to determine which treatment is acceptable in terms of color, aroma, taste, flavor, texture, and general acceptability using the 9-point Hedonic Scale. The quality scoring method was used to describe the attributes of the finished-product (Hilapad et. Al, 2020).

Physico-Chemical Analysis of Sweetpotato as Extender for Meat Loaf

The pH and Total Soluble Solid (TSS) of Meat Loaf as affected by varying levels of sweetpotato was determined with the use of a pH meter and a handheld refractometer, respectively.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 9

Color Analysis of Meat Loaf as Affected by Varying Levels of Sweetpotato

The Hunter L^* , a^* , and b^* values of meat loaf were measured using a colorimeter. The dimension L^* means lightness, a^* value indicates redness when positive and greenness when negative and the b^* value represents yellowness when positive and blueness when negative.

Browning index was calculated following the formula of Mohapatra et al (2010):

$$BI = 100 \times \frac{X - 0.31}{0.17}$$

Where:

$$X = \frac{a^* + 1.75L^*}{(5.645L^* + a^* - 3.012b^*)}$$

Yellowness index was calculated following the formula:

$$YI = 142.86b^*/L^*$$

Consumer Acceptance Testing

The formulation that was found most acceptable among laboratory panelists was subjected to consumer testing employing one hundred (100) consumers composed of one hundred college students of Capiz State University Mambusao Satellite College.

Cost Analysis

Production costs were determined by recording all the expenses incurred per formulation. The Computation of prices of raw materials and ingredients is based on the time of the purchase and not on the current price.

Statistical Tools and Analysis

The results of the sensory evaluation were analyzed statistically using the Analysis of Variance (ANOVA) and the Duncan Multiple Range Test (DMRT) was employed to locate significant differences among treatments (Larmond, 1970).

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Sensory Evaluation

The meat loaf made from different treatments was subjected to sensory evaluation to determine its acceptability and description for each attribute and the data is presented in Table 1.0. Meat loaf containing 125 g sweetpotato consistently obtained the highest mean acceptability rating in terms of color, aroma, taste, texture, flavor and general acceptability while meat loaf without sweetpotato in the formulation obtained the lowest ratings in all parameters being evaluated. The quality descriptions of meat loaf were yellowish brown for color, slightly perceptible sweetpotato aroma to well-blended sweetpotato and meat aroma, just

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 6 of 9

right to slightly sweet taste, moderately soft and tender to just right texture and slightly perceptible sweetpotato flavor. The data suggest that as the level of sweetpotato increased, its acceptability also increased. Statistical analysis showed that meat loaf containing 125 grams of sweetpotato was significantly different from treatment without sweetpotato and up to 75 grams in the formulation.

Table 1.0 Summary of mean acceptability ratings of meat loaf as affected by varying levels of sweetpotato

Treatment	Levels of Sweetpotato (g)	Color	Aroma	Taste	Texture	Flavor	GA
0	0	6.79c	6.97c	6.84e	6.83b	6.83d	7.33c
1	25	6.90cd	7.17b	6.93d	7.09b	7.16bc	7.56b
2	50	7.00c	7.00c	7.10c	7.00b	7.12c	7.63b
3	75	7.33b	7.28b	7.19c	7.28a	7.29ab	7.83a
4	100	7.36a	7.22b	7.42b	7.34a	7.23b	7.89a
5	125	7.44a	7.49a	7.56a	7.49a	7.38a	7.93a

Means followed by the same letter are not significantly different from one another at $p \geq 0.01$

Physico-chemical Characteristics of meat loaf with varying levels of sweetpotato

Meat loaf containing 125 g sweetpotato obtained the highest Total Soluble Solids (TSS) content of 21.0 while treatment 1 which contained 25 g sweetpotato had the lowest TSS content of 9.5. The higher TSS was attributed to the amount of sweetpotato added in the formulation which was responsible for its sweet taste. However, meat loaf containing 125 g sweetpotato had the lowest pH of 6.67.

Table 2.0 Total Soluble Solids and pH of meat loaf as affected by varying levels of sweetpotato

TREATMENT	LEVELS OF SWEETPOTATO (g)	TOTAL SOLUBLE SOLIDS ^{ns}	pH ^{ns}
0	0 g	10.3	7.10
1	25 g	9.5	6.97
2	50 g	12.7	6.70
3	75 g	15.4	6.93
4	100 g	15.3	6.87
5	125 g	21.0	6.67

ns- not significantly different from one another at $p \geq 0.05$

Color Analysis

The Browning index and yellowness index of meat loaf as affected by varying levels of sweetpotato is presented in Table 3.0. The data shows that meat loaf formulation without sweetpotato had the highest browning index. The high browning index was due to the color of the meat which was slightly brown when cooked (Divatkal and Menderrata, 2004). Whereas, meat loaf formulation containing the highest amount of sweetpotato had the highest yellowness

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 9

index. The result showed that the yellow color of sweetpotato used in the study influenced the resulting color of the finished product. However, statistical analysis showed no significant difference among treatments.

Table 3.0 Browning index and yellowness index of meat loaf as affected by varying levels of sweetpotato

TREATMENT	LEVELS OF SWEETPOTATO	BROWNING INDEX ^{ns}	YELLOWNESS INDEX ^{ns}
0	0 g	120.2	70.0
1	25 g	67.2	85.7
2	50 g	112.0	68.4
3	75 g	98.5	89.3
4	100 g	111.3	92.3
5	125 g	32.0	91.8

ns- not significant at $p \geq 0.05$

Consumer Testing

Meat loaf containing 125 grams sweetpotato was subjected to consumer testing to determine its acceptability among consumers. Results revealed that 84% of the consumers liked the meat loaf while only 16% disliked the product.

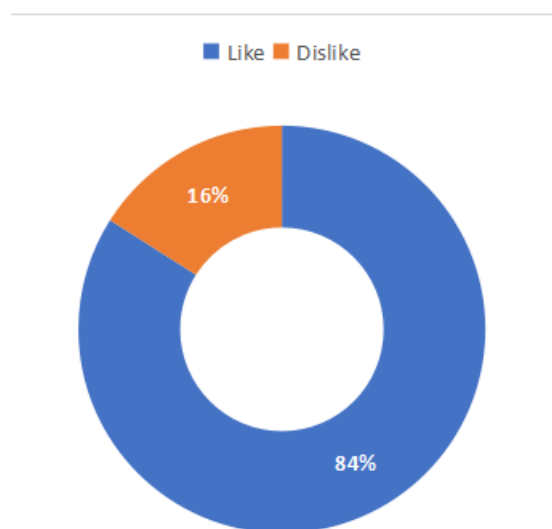


Figure 1. Consumer acceptability of meatloaf as affected by varying levels of sweetpotato

Cost Analysis

Table 4.0 shows the production cost of meat loaf from different formulations. The data revealed that the addition of sweetpotato in the formulation resulted in a decrease of its production cost by almost 50 percent.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 8 of 9

Table 4.0 Cost of production of the meat loaf as affected by varying levels of sweetpotato

Treatment	Level of Sweetpotato (g)	Cost per Formulation (PhP)
0	0 g	103.00
1	25 g	96.00
2	50 g	89.00
3	75 g	82.00
4	100 g	75.00
5	125 g	68.00

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the results of the study, researchers have drawn the following conclusions:

1. Meat loaf containing 125 grams of sweetpotato obtained the highest acceptability ratings in terms of color, aroma, taste, texture, flavor and general acceptability.
2. Meat loaf containing 125 grams of sweetpotato had the highest Total Soluble Solids Content and the lowest pH.
3. Meat loaf without sweetpotato had the highest browning index while meat loaf containing 125 grams of sweetpotato had the highest yellowness index.
4. Meat loaf containing 125 grams of sweetpotato was liked by 84% of the consumers and disliked by 16%.
5. Meat loaf containing 125 grams of sweetpotato had the lowest cost of production while those without sweetpotato had the highest production cost.

Recommendations

Based on the results of the study, the researchers recommended the following:

1. Use 125 grams sweetpotato in meat loaf formulation.
2. Conduct shelf-life testing of the most acceptable formulation.
3. Nutritional analysis of the most acceptable product is highly recommended.
4. Encourage farmers to plant more sweetpotatoes to have a steady supply of the commodity.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

- 1 drafted paper for publication
- 1 product
- 1 potential adopter
- 1 UM application for filing

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 9 of 9

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

The production cost of meat loaf from different formulations revealed that the addition of sweetpotato in the formulation resulted in the decrease of its production cost by almost 50 percent. Hence, the result of this study can help augment the income of farmers as the processing of this kind of product could result in increased profit. Additionally, farmers may either supply the raw materials or directly process the meatloaf for home consumption and for improved nutrition

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

The authors would like to thank Food Technology Department of Capiz State University Mambusao Satellite Campus for the support to this research endeavor.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

References

1. Cooper, T.L.; Smith, D.; Gonzales, M.J.C.; Maghanay, M.T.; Sanderson, S.; Cornejo, M.R.J.C.; Pineda, L.L.; Sagun, R.A.A.; Salvacion, O.P. Beyond numbers: Determining the socioeconomic and livelihood impacts of African swine fever and its control in the Philippines. *Front. Vet. Sci.* **2022**, *8*, 1678.
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